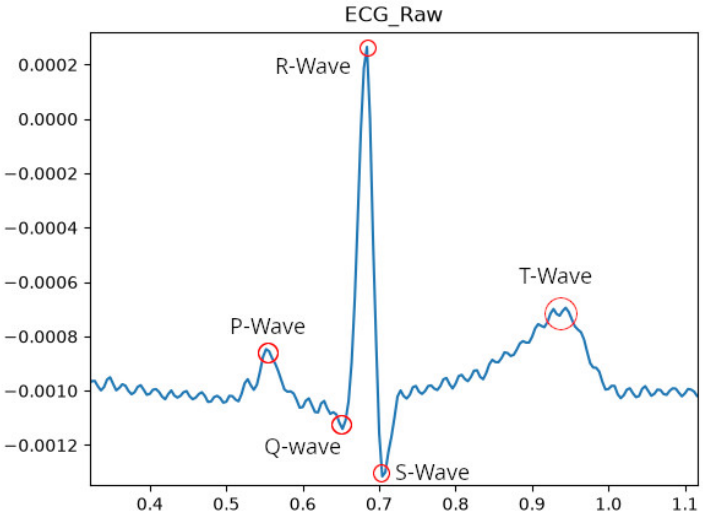
Digital Signal Processing
ECG Filtering Assignment

2785209M & 2664680S

0.1 Plotting the ECG Signal

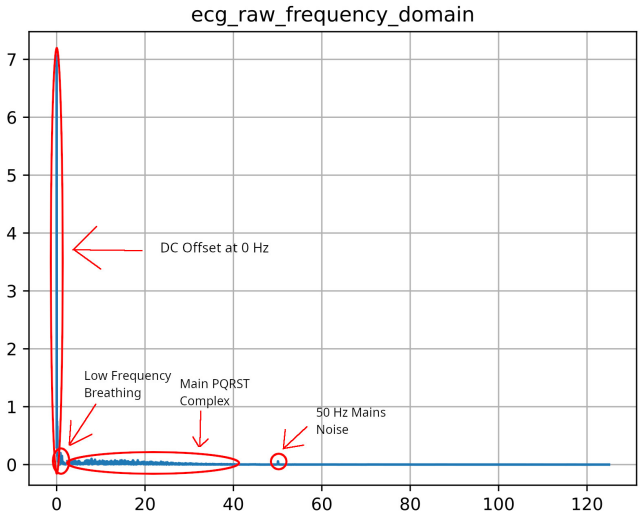
The experiment began by taking an ECG recording and plotting it in python using matplotlib. The electrodes were attached just above the hip and on the front of the shoulder.

0.1.1 Structure of PQRST Complex



It is important that we preserve the frequencies used to construct the PQRST complex, otherwise we would be destroying important data from the ECG.

0.1.2 Sources of Noise in the ECG



The main sources of noise in an ECG are 50Hz mains interference and random noise caused by movement and breathing. There is also a DC offset which shifts the entire signal by a certain amount, caused by potential difference formed between the electrodes and the skin.

0.2 Making the FIR Filter

0.2.1 calculating the number of coefficients and Stopband Attenuation

0.2.2 Performing IFFT on the ideal frequency response in the time domain

0.2.3 The Kaiser Window